

Coding & Solution

Date	30 September 2026
Team ID	6ab39579211fa4fd5d28901b
Project Name	EduGenie: Google Gemini Powered Learning Assistant
Maximum Marks	5 Marks

Coding & Solution

Solution Description:

EduGenie is implemented as a modular learning assistant that provides question answering, concept explanation, quiz generation, summarization, and personalized learning paths through a web interface.

Instructions:

1. The code is organized into separate feature modules and follows a consistent project structure.
2. The implementation addresses the five defined EduGenie learning features.
3. Source files, configuration, templates, static assets, requirements, README, and setup documentation are included in the repository.
4. The project repository is available on GitHub.

Solution Summary

Field	Details
Repository Link / URL	https://github.com/thamijudeen1706/EduGenie-Gemini-Learning-Assistant
Programming Language(s)	Python, HTML, CSS, JavaScript
Framework(s) Used	FastAPI, Uvicorn, Jinja2, Transformers, PyTorch, Google Gemini API
Key Features Implemented	Question Answering ,Concept Explanation, Quiz Generation, Summarization, Personalized Learning Path
Pending / Incomplete Features	None. All five defined core EduGenie features are implemented and functionally tested.
Setup / Run Instructions	Create and activate the Python virtual environment., Install dependencies: <code>pip install -r requirements.txt</code> , Configure the Gemini API key in .env: <code>GEMINI_API_KEY=YOUR_GEMINI_API_KEY</code> , Run the application: <code>python -m uvicorn main:app --reload</code> and Open: <code>http://127.0.0.1:8000/</code>

Code Quality Checklist

S.No	Criteria	Status (Yes / No)
1	Code is modular and organized into functions / classes	Yes
2	Meaningful variable and function names are used	Yes
3	Code includes comments / documentation where necessary	Yes
4	Error handling is implemented for critical operations	Yes
5	The application runs without critical errors	Yes
6	Code is committed to a version control repository	Yes

Additional Notes / Comments:

- The repository contains the complete EduGenie source code, configuration files, templates, static assets, requirements, README, and project documentation.
 - The five defined learning features were tested through the local FastAPI web application.
 - Gemini 3.5 Flash-Lite is used for Q&A, quiz generation, summarization, and personalized learning paths.
 - LaMini-Flan-T5-783M is used locally for concept explanation.
 - The application uses environment configuration for the Gemini API key; the secret key is not stored in the GitHub repository.
 - GitHub repository: <https://github.com/thamijudeen1706/EduGenie-Gemini-Learning-Assistant>
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